Ch. 25 CW Solutions

Principles of Macroecomonics

February 4, 2026

1. (a) Cost of Basket:

$$25(\$11) + 12(\$700) + 1(\$120) + 1(\$40) = \$8,835$$

Cost of Basket in Base Year:

$$25(\$10) + 12(\$600) + 1(\$100) + 1(\$50) = \$7,600$$

CPI:

$$\frac{\$8,835}{\$7,600} \cdot 100 = 116$$

(b)

$$\left(\frac{116 - 100}{100}\right) 100 = 16\%$$

(c) They are worse off because their nominal income rose less than the price level. That means their purchasing power decreased.

2. Real income in 1985:

$$\$23,618 \left(\frac{100}{107.6}\right) = \$21,949.81$$

Real income in 1995:

$$\$34,075 \left(\frac{100}{152.4}\right) = \$22,358.92$$

Real income in 2005:

$$\$46,326 \left(\frac{100}{195.3}\right) = \$23,720.43$$

Real income in 2015:

$$\$57,230 \left(\frac{100}{237}\right) = \$24,147.68$$

3. (a) Cost of basket in 2022:

$$30(\$4) + 10(\$8) + 10(\$8) = \$280$$

Cost of basket in 2023:

$$30(\$4) + 10(\$9) + 10(\$10) = \$310$$

Percent change:

$$\frac{310 - 280}{280} \cdot 100 = 10.7\%$$

(b) In 2023, two chickens cost less than one ham. As a result, consumers will buy zero hams and twenty additional chickens. Therefore, the cost of their actual consumption basket is

$$50(\$4) + 0(\$9) + 10(\$10) = \$300.$$

That means the actual percent change in the cost of consumption is

$$\frac{300 - 280}{280} \cdot 100 = 7.1\%,$$

so substitution bias made my answer to (a) 3.6 percentage points too high.

4. In 2025, Albert has

$$\$1,000(1.06)^3 = \$1,191.02.$$

In real dollars, that's

$$\$1,191.02 \left(\frac{100}{118}\right) = \$1,009.34.$$

Therefore, his purchasing power increased

$$\frac{\$1,009.34 - \$1,000}{\$1,000} \cdot 100 = 1\%.$$

5. They expect inflation in year one to be

$$\frac{110 - 100}{100} \cdot 100 = 10\%,$$

and they expect inflation in year two to be

$$\frac{121 - 110}{110} \cdot 100 = 10\%.$$

For Frank to get a real annual interest rate of 2%, he needs to charge a nominal annual interest rate of

$$2 + 10 = 12\%.$$

6. Let's say that this year is our base year, so today's CPI is 100. Then, the CPI in 16 years is $100(1.06)^16 = 254.04$. That means her one-time payment will be worth

$$\$20,000\left(\frac{100}{254.04}\right) = \$7,872.93$$

7. The real value of a home in 1963 was

$$\$18,000\left(\frac{100}{30.6}\right) = \$58,823.53,$$

and in 2025, it was

$$\$426,800\left(\frac{100}{322}\right) = \$132,546.60.$$

Therefore, the average real rate of return on homeownership was

$$\left(\left(\frac{\$132,546.60}{\$58,823.53}\right)^{\frac{1}{2025-1963}} - 1\right) \cdot 100 = 1.32\%$$